METRIC

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SUPERSEDING
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2 February 1993

MILITARY SPECIFICATION

CARTRIDGE, CALIBER .50, BLANK, XM928,

PERFORMANCE SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, and preparation for delivery for Cartridge, Caliber .50, XM928 Blank for use in Caliber .50 M2 Browning Machine Gun With M19 Blank Firing Attachment (BFA) and Caliber .50 M85 MG with M20 BFA. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305
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SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Untreated
O-I-503	- Insect Repellant
P-D-680	- Dry Cleaning Solvent
VV-L-800	- Lubricating Oil, General Purpose Preservative, (Water-Displacing, Low Temperature)

MILITARY

MIL-C-372	- Cleaning Compound Solvent for Bore of Small Arms and Automatic Aircraft Weapons
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-A-2550	- Ammunition, General Specification for
MIL-L-14107	- Lubricating Oil, Weapons Low Temperature
MIL-L-46000	- Lubricant, Semi-Fluid (Automatic Weapons)
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-L-63460	- Lubricant, Cleaner and Preservative for Weapons and Weapon Systems

STANDARDS

MILITARY

DOD-STD-100	- Engineering Drawing Practices
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking for Small Arms Ammunition
MIL-STD-650	- Explosives: Sampling, Inspection, and Testing
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-973	- Configuration Management
MIL-STD-1168	- Ammunition Lot Numbering

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

*	7046650	- Mount Machine Gun [Pintle and Ammo Tray Assembly used on M66 Ring Mount and M4 Pedestal Mount]
	7553296	- Box, Ammunition, M2A1 Assembly
	7553347	- Box, Wirebound Boxes, Ammunition, M2A1
	7643674	- Classification of Cartridge Case Defects Small Arms Ammunition
*	8648301	- Gage, Receiver, Profile and Alignment, Caliber .50 Cartridges
*	8648303	- Gage, Plug, Acceptance Check Shoulder Angle
*	8648306	- Gage, Plug, Wear Limit Check - Neck
*	8648903	- Gage, Fixture, Indicating - Head Dia.
*	8648904	- Gage, Fixture, Indicating - Diameter of Extractor Groove
*	10885124	- Installation, Pedestal - Gun Mount, M4
	12576456	- Packing and Marking for Box, Wirebound for Cartridge, Caliber .50
	12576457	- Packing and Marking for Box, Ammunition, M2A1 for Cartridge, Caliber .50
	12576458	- Cartridge, Caliber .50 Blank, XM928
*	EL8649403	- Screen Perforation
*	EL8649404	- Muzzle Flash
*	EL8649405	- Fouling
*	EL8649406	- Function and Casualty
*	EL8649407	- Noise Level
*	EL8654131	- Residual Stress
*	EL8654133	- Waterproofness
*	EL8658292	- Primer Sensitivity
	19-48-4116/14	- Unitization Procedures for Boxed Ammunition and Components on Four Way Entry Pallets

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PUBLICATIONS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

TB-700-2	-	Department of Defense Explosives Hazard Classification Procedures
TECP 700-700 VOL. III	-	Manual of Test Methods for Small Arms Ammunition
JSSPM	-	Joint Services Safety and Performance Manual for Qualification of Explosives for Military Use

TEST OPERATIONS PROCEDURES (TOPS)

U.S. ARMY TEST AND EVALUATION COMMAND (TECOM)

TOP 1-2-601	-	Laboratory Vibration Schedule
TOP 3-2-045	-	Automatic Weapons, Machine Guns, Hand and Shoulder Weapons
TOP 4-2–601	-	Drop Tower Tests for Munitions
TOP 4-2-602	-	Rough Handling

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.
Orders should specify, "49 CFR 100-199 (LATEST REVISION")."

(Copies of other Government documents, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Nongovernment publications. The following document(s) forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the-documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

American National Standards Institute (ANSI)

ANSI Y14.5M-1982 - Dimensioning and Tolerancing

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 General. The cartridges shall comply with the requirements shown on drawing 12576458, reference specifications and the requirements stated in this section. All of the cartridge components shall be compatible with each other. The primer mix shall be noncorrosive and the propellant shall be smokeless. The cartridge shall be compatible with Caliber .50 M9 and M15A2 links, the M2 Browning Machine Gun (M2BMG) with M19 BFA, and M85 Machine Gun with M20 BFA weapon systems. When ammunition is to be procured for use with M2 Browning Machine Guns the cartridges shall be linked with M9 links. When ammunition is to be procured for use with M85 Machine Guns the cartridges shall be linked with M15A2 links.

3.2 Qualification. A sample shall be submitted for qualification in accordance with (IAW) the technical provisions specified herein (see section 4). The purpose of qualification is to verify that a particular contractor's design shall be considered qualified indefinitely. The contractor shall submit level 3 detailed item drawings IAW DOD-STD-100, MIL-A-2550 and ANSI Y14.5-82 to the Government for configuration control. All changes shall be submitted IAW MIL-STD-973 for review and approval by the Government. The Government reserves the right to subject the revised design to any or all the qualification evaluations specified in 4.3 at the expense of the contractor.

3.3 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection IAW the technical provisions specified herein (see 4.4).

3.4 Performance characteristics.

3.4.1 Primer sensitivity. The value of the average height (h) minus 2 standard deviations (s) shall be equal to or more than 63mm (2.5"). For cartridges with metal cases the value of h + 5s shall be equal to or less than 381mm (15"). For cartridge cases made of other materials, the value of h + 5s shall be established by the contractor prior to completion of Qualification testing.

3.4.2 Residual stress. The cartridge shall not split when subjected to the accelerated stress test for the applicable materials.

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3.4.3 Waterproof. The cartridge shall release not more than one bubble of air when tested at a pressure of 14 kPa (2 psi) below atmospheric pressure for 15 seconds.

3.4.4 Function and casualty. The cartridges shall function without casualty throughout a temperature range of -20C (-4F) to +50C (122F). The cartridges shall safely function throughout a temperature range of -50C (-58F) through +70C (158F).

3.4.4.1 Cyclic rate.

3.4.4.1.1 M2 BMG. The blank cartridges when temperature conditioned shall operate the M2 BMG with M19 BFA at an average cyclic rate between 450 and 600 shots per minute (spin) throughout the temperature range of -20C (-4F) to +50C (122F).

3.4.4.1.2 M85 MG. The blank cartridges when temperature conditioned shall operate the M85 MG with M20 BFA at an average cyclic rate between 450 and 600 shots per minute (spin) throughout the temperature range of -20C (-4F) to +50C (122F).

3.4.4.2 Noise level. The noise levels of cartridges shall be no greater than 164 decibels nor more than 10 decibels below the noise level of M33 ball cartridges fired under identical conditions.

3.4.4.3 Screen perforation. The cartridges conditioned throughout the temperature range of -20C (-4F) to +50C (122F) when fired in a M2 BMG with M19 BFA or M85 MG with M20 BFA, shall not cause perforations greater than or equal to 2.5mm (.1 inch) diameter in a paper screen placed perpendicular to the line of fire at a distance of 5 meters (15 feet) from the muzzle of the weapon.

3.4.4.4 Muzzle flash. The flash produced at the muzzle of both the M2 BMG with M19 BFA and M85 MG with M20 BFA shall be visible at 100 meters (110 yards) from the muzzle of the weapon in the dark.

3.4.4.5 Breech flash. Breech flash when firing cartridges shall not be unsafe for the gunner.

3.4.4.6 Fouling. The fouling produced by firing 500 cartridges shall not cause more than one weapon stoppage, any safety problems, and shall be readily removable using standard cleaning methods. The fouling produced by firing 500 cartridges shall not cause a change in cyclic rate from the first 100 rounds to the final 100 rounds greater than 15% when fired in M2 BMG with M19 BFA and M85 MG with M20 BFA .

* 3.4.4.7 Attitude: The cartridge shall operate the M2 BMG with M19 BFA when fired at an attitude of ± 85 degrees.

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3.4.4.8 Belt pull. The cartridge shall operate the M2 BMG with M19 BFA and M85 MG with M20 BFA when a 100 round belt of cartridges is loaded in the weapon and the belt of cartridges is allowed-to hang freely.

3.4.4.9 Hot chamber effect. A cartridge inserted in the chamber immediately following the continuous firing of 100 cartridges shall not fire (cook off). In addition the cartridge shall not deform to the extent that clearing of the weapon is made difficult following the proper waiting period.

3.4.4.10 Chemical compatibility. The reliability of the cartridge shall not be less than .900 at 90% confidence level after having been immersed in the following lubricants solvents, or chemicals:

Specification	Material
MIL-L-46000	Lubricant, Semi-Fluid (LSA)
MIL-L-14107	Lubricating Oil, (LAW)
MIL-L-2104	Lubricating Oil, (ICE, T)
MIL-L-46167	Lubricating Oil, (ICE, A)
MIL-L-63460	Lubricant, (CLP)
VV-L-800	Lubricating Oil, (PL-S)
MIL-C-372	Bore Cleaner, (RBC)
O-I-503	Insect Repellant
P-D-680	Dry Cleaning Solvent (SD-1)

3.5 Environmental.

3.5.1 Sand and dust. The reliability of the cartridge shall not be less than .900 at 90% confidence level of the reliability of the Caliber .50 M33 Ball cartridges when both are subjected to the sand and dust test.

3.5.2 Corrosion. The cartridge shall fire safely in the weapon after being subjected to a 5% salt spray solution for 48 hours.

3.5.3 Temperature/humidity cycling. After being subjected to a 10 day temperature/humidity cycle, the cartridge when fired shall not change the cyclic rate of the weapon by more than 15% and the reliability shall not be less than .900 at 90% confidence level.

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3.5.4 Long term storage.

3.5.4.1 Long term high temperature. After being subjected to 30 days of continuous heating at 65C (150F) and 10 to 20% relative-humidity, the cartridge when fired shall not change the cyclic rate of the weapon by more than 20%. The reliability shall not be less than .900 at 90% confidence level. If the cartridge fails to meet either of these requirements, the cartridge shall be retested after conditioning at 50C (122F) and 10 to 20% relative humidity for 120 days.

3.5.4.2 Long term low temperature. After being subjected to 30 days of continuous cooling at -40C (-40F), the cartridge when fired shall not change the cyclic rate of the weapon by more than 20%. The reliability shall not be less than .900 at 90% confidence level.

3.5.5 Rough handling.

3.5.5.1 Sequential. Cartridges when packaged in wirebound boxes, shall be subjected to the transportation-vibration, loose cargo, 1.5 meter drop test, and the 2.1 meter drop tests. The cartridges shall be safe to handle, store, and fire. The reliability shall not be less than .900 at 90% confidence level.

3.5.5.2 12 meter drop. After being subjected to the 12 meter (40 ft) drop test the wirebound packaged cartridges shall not detonate, deflagrate, loose propellant, and must be safe to handle and to dispose.

3.5.6 Toxic fumes. The toxic fumes produced by the cartridge shall be acceptable to the Surgeon General for the applicable training scenarios.

3.6 Safety.

3.6.1 Safety hazard classification. The safety hazard classification for the cartridge packed for shipping and storage shall be:

DOD HAZARD CLASS/DIV/SCG: 1.4C
DOT HAZARD CLASS: 1.4C
DOT PROPER SHIPPING NAME: CARTRIDGES SMALL ARMS BLANK
UN SERIAL NUMBER: 0338

3.6.2 Net explosive weight. The net explosive weight of the cartridge (sum of the propellant, primer, and any other pyrotechnic material shall not exceed 15 grams (230 grains).

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3.6.3 Explosive sensitivity. The contractor shall provide explosive sensitivity data on all energetic material contained in the cartridges. The sensitivity for the materials shall be as indicated below:

Sensitivity	Requirement
a. Impact, ERL Type 12 tools 2.4 kg, cm @ 50% point	20 cm (min)
b. Friction Pendulum	
- Steel Shoe	Burn - no detonation
- Fiber Shoe	Burn - no detonation
c. Electrostatic Discharge .030 Joule	No detonation in 20 tests
d. Differential Thermal Analysis	200 C (min)

3.7 Workmanship. All plastic parts shall be fabricated in a thorough workmanlike manner. Plastic parts shall be free of porosity, twist, warpage, heat marks, sinks, voids, cracks, chipped edges, burrs, bubbles, blisters, burn marks, or other defects that will affect serviceability. All metal parts shall be fabricated in a thorough workmanlike manner. Metal parts shall be free of burrs, chips, chipped edges, gouges, cracks, splits, surface defects, dust, dirt, grease, oil, lacquer smears, and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by any cleaning agent. All assemblies shall be assembled in a thorough workmanlike manner. Extreme care should be exercised to avoid contamination of primers or propellant by oil, grease, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's

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overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Refer to MIL-STD-109 for definitions of quality assurance terms used.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification (See 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).

4.3 Qualification.

4.3.1 Submission. The contractor shall submit Qualification sample of 36,000 cartridges for evaluation IAW the provisions of 4.3.3.

4.3.2 Qualification sample. The qualification sample shall be made to the same design as will be submitted for production. The qualification sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used for production. Where production equipment is not available, equipment that is representative of the production equipment shall be used. The qualification sample shall not be custom made in such a way that the product or product quality will be different from production ammunition.

4.3.3 Inspection to be performed. As determined by the Government, the qualification assemblies, components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (See MIL-A-48078 and Tables I and II) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.4 Rejection. See MIL-A-48078 and Tables I and II herein. Submission of a qualification sample that is not representative of normal-production shall be cause for rejection of that sample and denial of design qualification. Failure of the qualification sample to be IAW the applicable drawings and the requirements of Table I and II, as specified herein, will result in disapproval of the qualification sample and denial of qualification of the submitted design.

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TABLE I. Qualification.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 Blank, XM928 12576458

EXAMINATION OR TEST	SAMPLE SIZE	CRITERIA AC	CRITERIA RE	REQUIREMENT PARAGRAPH	INSPECTION REFERENCE
Examination for defects 1/					
Major Class	800	2	3	3.1	4.5.2.1
Minor Class	800	10	11	3.1	4.5.2.1
Primer sensitivity	500	2/		3.4.1	4.6.1.1
Residual stress	125	2	3	3.4.2	4.6.1.2
Waterproof	200	10	11	3.4.3	4.6.1.3
M2 BMG with M19 BFA on M3 Tripod Mount					
Function and casualty at 20C	1000	TABLE II		3.4.4	4.6.1.4
Cyclic rate	1000	2/		3.4.4.1	4.6.1.4.1
M2 BMG with M19 BFA on MK93 DUAL MOUNT (on HMMWV)					
Function and casualty at 20C	1000	TABLE II		3.4.4	4.6.1.4
Cyclic rate	1000	2/		3.4.4.1	4.6.1.4.1
M2 BMG with M19 BFA on M66 RING MOUNT (on 5 TON TRUCK)					
Function and casualty at 20C	1000	TABLE II		3.4.4	4.6.1.4
Cyclic rate	1000	2/		3.4.4.1	4.6.1.4.1
M2 BMG with M19 BFA ON HARD MOUNT					
Function and casualty					
a. 20C	3000	TABLE II		3.4.4	4.6.1.4
b. -20C 4/	3000	TABLE II		3.4.4	4.6.1.4
c. 50C 4/	3000	TABLE II		3.4.4	4.6.1.4
d. -50C 4/	200	6/		3.4.4	4.5.1.4
e. 70C 4/	200	6/		3.4.4	4.5.1.4
Cyclic rate					
a. 20C	1000	2/		3.4.4.1	4.6.1.4.1
b. -20C 4/	1000	2/		3.4.4.1	4.6.1.4.1
c. 50C 4/	1000	2/		3.4.4.1	4.6.1.4.1
Noise level 20C 3/	100	2/		3.4.4.2	4.6.1.4.2
Screen perforation 3/					
a. 20C	300	2	3	3.4.4.3	4.6.1.4.3
b. -20C 4/	300	2	3	3.4.4.3	4.6.1.4.3
c. 50C 4/	300	2	3	3.4.4.3	4.6.1.4.3
Muzzle flash (20C)	20	0	1	3.4.4.4	4.6.1.4.4
Breech flash	4/	0	1	3.4.4.5	4.6.1.4.5
Fouling 20C 3/	500	2/		3.4.4.6	4.6.1.4.6
Attitude					
a. +85 degrees elevation	100	2/		3.4.4.7	4.6.1.4.7
a. -85 degrees elevation	100	2/		3.4.4.7	4.6.1.4.7
Belt pull	100	2/		3.4.4.8	4.6.1.4.8
Hot chamber affects	303	2/		3.4.4.9	4.6.1.4.9
Chemical compatibility				3.4.4.10	4.6.1.4.10
Lubricant, Semi-Fluid (LSA)	22	7/			
Lubricating Oil, (LAW)	22	7/			
Lubricating Oil, (ICE, T)	22	7/			

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TABLE I. (Cont.) Qualification.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 Blank, XM928 12576458

EXAMINATION OR TEST	SAMPLE SIZE	CRITERIA AC	CRITERIA RE	REQUIREMENT PARAGRAPH	INSPECTION REFERENCE
M2 BMG with M19 BFA ON HARD MOUNT (Cont.)					
Chemical compatibility (Cont.)				3.4.4.10	4.6.1.4.10
Lubricating Oil, (ICE, A)	22	7/			
Lubricant, (CLP)	22	7/			
Lubricating Oil, (PL-S)	22	7/			
Bore Cleaner, (RBC)	22	7/			
Insect Repellant	22	7/			
Dry Cleaning Solvent (SD-1)	22	7/			
Environmental					
Sand and Dust	200	14	15	3.5.1	4.6.2.1
Corrosion	10	2/		3.5.2	4.6.2.2
Temperature/humidity cycling	600	50	51	3.5.3	4.6.2.3
Long term storage					
High temperature	200	14	15	3.5.4.1	4.6.2.4
Low temperature	200	14	15	3.5.4.2	4.6.2.4
Rough handling					
Transport vibration & drop	400	32	33	3.5.5.1	4.6.2.5.1
12 meter drop	400	2/		3.5.5.2	4.6.2.5.2
Toxic fumes	300	2/		3.5.6	4.6.2.6
M85 Machine Gun with M20 BFA ON HARD MOUNT					
Function and casualty					
a. 20C	3000	TABLE II		3.4.4	4.6.1.4
b. -20C 4/	3000	TABLE II		3.4.4	4.6.1.4
c. 50C 4/	3000	TABLE II		3.4.4	4.6.1.4
d. -50C 4/	200	6/		3.4.4	4.5.1.4
e. 70C 4/	200	6/		3.4.4	4.5.1.4
Cyclic rate					
a. 20C	1000	2/		3.4.4.1	4.6.1.4.1
b. -20C 4/	1000	2/		3.4.4.1	4.6.1.4.1
c. 50C 4/	1000	2/		3.4.4.1	4.6.1.4.1
Noise level 20C 3/	100	2/		3.4.4.2	4.6.1.4.2
Screen perforation 3/					
a. 20C	300	2	3	3.4.4.3	4.6.1.4.3
b. -20C 4/	300	2	3	3.4.4.3	4.6.1.4.3
c. 50C 4/	300	2	3	3.4.4.3	4.6.1.4.3
Muzzle flash (20C)	20	0	1	3.4.4.4	4.6.1.4.4
Breech flash	5/	0	1	3.4.4.5	4.6.1.4.5
Fouling 20C 3/	500	2/		3.4.4.6	4.6.1.4.6
Belt pull	100	2/		3.4.4.8	4.6.1.4.8
Hot chamber affects	303	2/		3.4.4.9	4.6.1.4.9

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TABLE I. (Cont.) Qualification.

CLASSIFICATION OF CHARACTERISTICS

NOTES: 1/ The examinations and the class, Major or Minor, shall be as shown in 4.5.2.1. All 800 cartridges are to be inspected for both Major and Minor characteristics. The levels of inspection shown do not apply to Qualification examinations.

2/ Failure of cartridges to meet the applicable requirement shall be cause for rejection of qualification.

3/ May be performed during function and casualty testing.

4/ The weapons shall be conditioned with the test ammunition.

5/ Observe during function and casualty testing.

6/ Extreme temperature tests are for safety evaluation only. Any safety problem caused by the cartridge shall be cause for rejection of qualification.

7/ Accept the qualification lot if there are no stoppages attributable to the ammunition. Reject the qualification lot if there are six or more stoppages attributable to the ammunition. If there are one to five stoppages attributable to the ammunition select a second sample of 71 additional cartridges which shall be fired. The qualification lot shall be accepted if no more than five stoppages attributable to the ammunition occur in firing the the total of 93 cartridges. Reject the qualification lot if there are six or more stoppages attributable to the ammunition.

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TABLE II. Qualification inspection - function and casualty firing defect acceptance - rejection criteria.

	Defect Description	Acceptance Number	Rejection 4/ Number
1.	Misfire	3	4
2.	Weapon stoppage attributable to ammunition 1/	35	36
3.	Separation of case components (when applicable)		
	a. Partial	43	44
	b. Complete	0	1
4.	Gas leaks between case components (when applicable)	17	18
5.	Failure to extract	3	4
6.	Primer defects:		
	a. Perforation in the primer cup caused by the firing pin	123	124
	b. Escape of gas around the primer cup (excluding a. above)	123	124
	c. Blown primer 2/	0	1
	d. Dropped primer 2/	3	4
	e. Escape of gas around the primer cup more than 50% of the periphery	43	44
7.	Case casualties 3/		
	a. Longitudinal split		
	(1) Neck (I)	143	144
	(2) Body (K)	0	1
	(3) Body (J)	43	44
	(4) To Head (L)	0	1
	(5) Through head (M)	0	1
	(6) Shoulder (S)	43	44
	b. Circumferential rupture		
	(1) Partial - neck, shoulder, or body (S, J, or K)	22	23
	(2) Partial - case base (L)	0	1
	(3) Complete	0	1
8.	Case burn through	0	1

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TABLE II. (Cont.) Qualification inspection - function and casualty firing defect acceptance - rejection criteria.

NOTES:

1/ Occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall it be recorded as a defect chargeable to the ammunition. All other stoppages attributable to the ammunition, with the exception of misfire, complete circumferential rupture, or failure to extract, shall be included in this defect category.

2/ Primer defect descriptions:

a. Blown primer. A primer which, when the cartridge is fired is separated completely from the head of the case and the primer pocket are grossly distorted and deformed. The severity of this condition is such that it is readily detectable by visual examination.

b. Dropped primer. A primer which falls from the primer pocket after the cartridge is fired.

3/ For location of defects indicated in parenthesis, see drawing 7643674.

4/ If the function and casualty defects plus firing defects observed in all other tests exceed the acceptance number, the qualification sample shall be rejected.

****

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation IAW provisions of 4.4.2. The first article sample shall consist of the following items in the quantities indicated:

Part Description	Drawing	Quantity
Cartridge, Cal.50	12576458	3250
Cases with Primers		500

4.4.2 Inspection to be performed. As determined by the Government, the first article assemblies, components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (See Table IV and MIL-A-48078) and be inspected for compliance with any or all requirements of the applicable drawings.

4.4.3 Rejection. See MIL-A-48078. Failure of the First Article sample to be IAW the applicable drawings and requirements of Table III. and Table IV. as specified herein, will result in disapproval of the First Article.

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TABLE III. First article inspection

Cartridge, Caliber .50 Blank, XM928 12576458

EXAMINATION OR TEST INSPECTION REFERENCE	SAMPLE SIZE	CRITERIA AC	RE	REQUIREMENT PARAGRAPH	
Examination for defects 1/					
Major Class	800	3	4	3.1	4.5.2.1
Minor Class	800	14	15	3.1	4.5.2.1
Primer sensitivity	500	2/		3.4.1	4.6.1.1
Residual stress	20	0	1	3.4.2	4.6.1.2
Waterproof	20	1	2	3.4.3	4.6.1.3
M2 BMG with M19 BFA ON HARD MOUNT					
Function and casualty					
a. 20C	300	TABLE IV		3.4.4	4.6.1.4
b. -20C	300	TABLE IV		3.4.4	4.6.1.4
c. 50C	300	TABLE IV		3.4.4	4.6.1.4
Cyclic rate					
a. 20C	300	2/		3.4.4.1	4.6.1.4.1
b. -20C	300	2/		3.4.4.1	4.6.1.4.1
c. 50C	300	2/		3.4.4.1	4.6.1.4.1
Noise level 20C 3/	100	2/		3.4.4.2	4.6.1.4.2
Screen perforation 3/					
a. 20C	200	2	3	3.4.4.3	4.6.1.4.3
b. -20C	200	2	3	3.4.4.3	4.6.1.4.3
c. 50C	200	2	3	3.4.4.3	4.6.1.4.3
Muzzle flash (20C)	20	0	1	3.4*4.4	4.6.1.4.4
Fouling 20C 3/	500	2/		3.4.4.6	4.6.1.4.6
M85 Machine Gun with M20 BFA ON HARD MOUNT					
Function and casualty					
a. 20C	300	TABLE IV		3*4.4	4.6.1.4
b. -20C	300	TABLE IV		3.4.4	4.6.1.4
c. 50C	300	TABLE IV		3.4.4	4.6.1.4
Cyclic rate					
a. 20C	300	2/		3.4.4.1	4.6.1.4.1
b. -20C	300	2/		3.4.4.1	4.6.1.4.1
50C	300	2/		3.4.4.1	4.6.1.4.1
Noise level 20C 3/	100	2/		3.4.4.2	4.6.1.4.2
Screen perforation 3/					
a. 20C	200	2	3	3.404.3	4.6.1.4.3
b. -20C	200	2	3	3.4.4.3	4.6.1.4.3
c. 50C	200	2	3	3.4.4.3	4.6.1.4.3
Muzzle flash (20C)	20	0	1	3.4.4.4	4.6.1.4.4
Fouling 20C 3/	500	2/		3.4.4.6	4.6.1.4.6

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TABLE III. (Cont.) First article inspection

Cartridge, Caliber .50 Blank, XM928	12576458

NOTES:

1/ The examinations and the class, Major or Minor, shall be as shown in 4.5.2.1. All 800 cartridges are to be inspected for both Major and Minor characteristics. The levels of inspection shown do not apply to First Article Inspection examinations.

2/ Failure of cartridges to meet the applicable requirement shall be cause for rejection of First Article.

3/ May be performed during function and casualty testing. For the fouling test an additional 200 rounds must be fired at 20C without cleaning through each machine gun for a total of 500 rounds fired through each machine gun.

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TABLE IV. First article inspection - function and casualty firing defect acceptance - rejection criteria.

Defect Description	Acceptance Number	Rejection Number
1. Misfire	1	2
2. Weapon stoppage attributable to ammunition 1/	2	3
3. Separation of case components (when applicable)		
a. Partial	13	14
b. Complete	0	1
4. Gas leaks between case components (when applicable)	5	6
5. Failure to extract	0	1
6. Primer-defects:		
a. Perforation in the primer cup caused by the firing pin	24	25
b. Escape of gas around the primer cup (excluding a. above)	7	8
c. Blown primer	0	1
d. Dropped primer	0	1
e. Escape of gas around the primer cup more than 50% of the periphery	14	15
7. Case casualties 2/		
a. Longitudinal split		
(1) Neck (I)	44	45
(2) Body (K)	0	1
(3) Body (J)	13	14
(4) To Head (L)	0	1
(5) Through head (M)	0	1
(6) Shoulder (S)	13	14
b. Circumferential rupture		
(1) Partial - neck, shoulder, or body (S, J, or K)	2	3
(2) Partial - case base (L)	0	1
(3) Complete	0	1
8. Case burn through	0	1

NOTES 1/ Occasionally the last round of a continuous belt due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall it be recorded as a defect chargeable to the ammunition. All other stoppages attributable to the ammunition, with the exception of misfire, complete circumferential rupture, or failure to extract, shall be included in this defect category.

2/ For location of defects indicated in parenthesis, see drawing 7643674.

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4.5 Quality conformance inspection.

4.5.1 Inspection lot formation. IAW MIL-A-48078.

4.5.1.1 Component parts. Unless otherwise specified, component parts shall be homogeneous and of a size convenient to the contractor and inspected, tested, and accepted by the contractor. The cartridge lot shall contain:

a. Cartridge cases from one manufacturer.
b. Primers from one interfix and not more than 2 lots.
c. Propellant from one interfix and not more than 2 lots.

4.5.1.2 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, lot number IAW MIL-STD-1168, and the supplier's identification as assigned by the procuring activity.

4.5.1.3 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.5.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein attributes sampling inspection shall be conducted IAW Table V below using the inspection levels cited in the Classification of Characteristics paragraphs.

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TABLE V. Attributes sampling inspection

	Inspection Levels	
	III	VI
LOT SIZE	SAMPLE SIZE	
TO 150	32	20
151 TO 280	32	20
281 TO 500	50	32
501 TO 1200	80	32
1201 TO 3200	80	32
3201 TO 10000	125	50
10001 TO 35000	125	50
35001 TO 150000	125	80
150001 TO 500000	200	80
500001 AND OVER	200	80

If the sample size exceeds the lot size, perform 100% inspection. Accept with zero defects only and reject with one or more defects for all inspection levels.

b. Alternative inspection provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment such as statistical process control, tool control, or other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.8). When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

4.5.2.1 Cartridge, Caliber .50 Blank, XM928 12576458

EXAMINATION	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION REFERENCE
CRITICAL: NONE DEFINED			
MAJOR: 1/			
101. CASE SPLIT (6) 2/	100%	3.1	VISUAL
102. PERFORATED CASE (7)	100%	3.1	VISUAL
103. PRIMER MISSING (32)	100%	3.1	VISUAL
104. PRIMER COCKED (33)	100%	3.1	VISUAL
105. PRIMER INVERTED (34)	100%	3.1	VISUAL
106. CARTRIDGE WEIGHT OVER MAX 3/	100%	3.1	GAGE
107. CORRODED OR STAINED (IF ETCHED) CASE (2)	LEVEL III	3.1	VISUAL
108. ROUND HEAD (CASE) (4)	LEVEL III	3.1	VISUAL
109. DENTED CASE (5)	LEVEL III	3.1	VISUAL
110. DRAW SCRATCH IN CASE (8)	LEVEL III	3.1	VISUAL
111. BEVELED UNDERSIDE OF HEAD (10)	LEVEL III	3.1	VISUAL
112. sCALY METAL (CASE) (12)	LEVEL III	3.1	VISUAL
113. NO CHAMFER ON HEAD (RIM) (13)	LEVEL III	3.1	VISUAL
114. LOOSE PRIMER (35)	LEVEL III	3.1	VISUAL/MANUAL
*115. CARTRIDGE PROFILE FAILURE 4/	LEVEL III	3.1	8648301 8648303 8648306
*116 DIAMETER OF EXTRACTOR GROOVE, MAXIMUM	LEVEL III	3.1	8648904
*117. DIAMETER OF HEAD, MAXIMUM	LEVEL III	3.1	8648903
118. THICKNESS OF RIM, MAXIMUM	LEVEL III	3.1	GAGE
119. DISTANCE TO SHOULDER DATUM	LEVEL III	3.1	GAGE
120. DEPTH OF PRIMER	LEVEL III	3.1	GAGE
121. PRIMER CUP MISSING	LEVEL III	3.1	VISUAL
122. OVERALL LENGTH, MAXIMUM	LEVEL III	3.1	GAGE

NOTE:

1/ Refer to visual defect standards in MIL-STD-636 for location of defects indicated in parenthesis.
2/ Regardless of location or occurrence of loss of propellant.
3/ The contractor may have this inspection requirement waived upon successful demonstration that a case 100% filled with propellant does not constitute a safety hazard.
4/ Requiring more than 355 N (80 lbf) to insert in profile and alignment gage.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

4.5.2.1 (Cont.) Cartridge, Caliber .50 Blank, XM928 12576458

EXAMINATION	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION REFERENCE
MINOR:			
201. DISCOLORED, DIRTY, OIL, OR SMEARED (WATERPROOFING)	LEVEL VI	3.1	VISUAL
202. DENTED CASE (5)	LEVEL VI	3.1	VISUAL
203. DRAW SCRATCH IN CASE (8)	LEVEL VI	3.1	VISUAL
204. SCRATCH IN CASE (9)	LEVEL VI	3.1	VISUAL
205. SCALY METAL ON CASE (12)	LEVEL VI	3.1	VISUAL
206. FOLD, WRINKLE BUCKLE, OR BULGE IN CASE (14, 15, 16, 17)	LEVEL VI	3.1	VISUAL
207. HEAD STAMP MISSING OR ILLEGIBLE (18)	LEVEL VI	3.1	VISUAL
208. DEFECTIVE HEAD (19)	LEVEL VI	3.1	VISUAL
209. NO WATERPROOFING MATERIAL (PRIMER POCKET JOINT) (37), IF APPLICABLE	LEVEL VI	3.1	VISUAL
210. DEFECTIVE CRIMP (PRIMER (38) IF APPLICABLE	LEVEL VI	3.1	VISUAL
211. WORKMANSHIP	LEVEL VI	3.1	VISUAL

1/ Refer to visual defect standards in MIL-STD-636 for location of defects indicated in parenthesis.

2/ Regardless of location or occurrence of loss of propellant.

3/ The contractor may have this inspection requirement waived upon successful demonstration that a case 100% filled with propellant does not constitute a safety hazard.

4/ Requiring more than 355 N (80 lbf) to insert in profile and alignment gage.

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4.5.3 Testing. Testing is described in the Quality Conformance Inspection Tables, paragraph 4.5.3.1. Testing shall be conducted IAW the methods and procedures specified in 4.6.

a. Test samples. The quantities for the various tests shall be as specified in 4.5.3.1. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. Sufficient cartridges shall be selected so that all testing (including testing of second samples where necessary) can be performed. The cartridges shall be thoroughly mixed before being divided into samples-for the various tests.

b. Firing defects. Firing defects and associated acceptance-rejection criteria are specified in Table VI.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

4.5.3.1 Cartridge, Caliber .50 Blank, XM928 12576458

TESTS	SAMPLE SIZE	CRITERIA AC	CRITERIA RE	REQUIREMENT PARAGRAPH	INSPECTION REFERENCE
Primer sensitivity	500	1/		3.4.1	4.6.1.1
Residual stress	20	2/		3.4.2	4.6.1.2
Waterproof	20	3/		3.4.3	4.6.1.3
M2 BMG with M19 BFA ON HARD MOUNT					
Function and casualty					
a. 20C	300	4/		3.4.4	4.6.1.4
b. -20C	300	4/		3.4.4	4.6.1.4
c. 50C	300	4/		3.4.4	4.6.1.4
Cyclic rate					
a. 20C	300	5/		3.4.4.1	4.6.1.4.1
b. -20C	300	5/		3.4.4.1	4.6.1.4.1
c. 50C	300	5/		3.4.4.1	4.6.1.4.1
Noise level 20C 6/	100	1/		3.4.4.2	4.6.1.4.2
Screen perforation 6/					
a. 20C	200	2	3	3.4.4.3	4.6.1.4.3
b. -20C	200	2	3	3.4.4.3	4.6.1.4.3
c. 50C	200	2	3	3.4.4.3	4.6.1.4.3
M85 Machine Gun with M20 BFA ON HARD MOUNT					
Function and casualty					
a. 20C	300			3.4.4	4.6.1.4
b. -20C	300			3.4.4	4.6.1.4
c. 50C	300			3.4.4	4.6.1.4
Cyclic rate					
a. 20C	300			3.4.4.1	4.6.1.4.1
b. -20C	300			3.4.4.1	4.6.1.4.1
c. 50C	300			3.4.4.1	4.6.1.4.1
Noise level 20C 6/	100			3.4.4.2	4.6.1.4.2
Screen perforation 6/					
a. 20C	200	2	3	3.4.4.3	4.6.1.4.3
b. -20C	200	2	3	3.4.4.3	4.6.1.4.3
c. 50C	200	2	3	3.4.4.3	4.6.1.4.3

NOTES: 1/ The lot shall be rejected if the requirement is not met.

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QUALITY CONFORMANCE INSPECTION

2/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test a second sample consisting of double the number of cartridges in the first sample shall be tested. If any additional cartridges fail in the second test the lot shall be rejected.

3/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, nine-or more cartridges fail to comply with the applicable requirement.

4/ See Table VI.

5/ The cyclic rate shall be measured during the function and casualty test. Failure of any 100 round burst to comply with the applicable requirement shall be cause for rejection of the lot, subject to testing a second sample consisting of double the number of cartridges in the first sample. Failure of any 100 round burst in the second sample to comply with the applicable requirement shall be cause for rejection of the lot. The second sample shall be fired in the same weapon type in which the failure occurred.

6/ These tests may be performed concurrently with function and casualty testing.

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TABLE VI

QUALITY CONFORMANCE INSPECTION
FUNCTION AND CASUALTY FIRING DEFECT CLASSIFICATION

Defect Description	Acceptance Numbers 1/ First Sample	Cumulative	Defect Category
1. Misfire	1	2	Major
2. Weapon stoppage attributable to ammunition	2	3	Major
3. Separation of case components (when applicable)			
a. Partial	1	3	Minor
b. Complete	0	0	Major
4. Gas leaks between case components (when applicable)	1	3	Minor
	0	1	Major
5. Failure to extract			
6. Primer defects:			
a. Perforation in the primer cup caused by the firing pin	24	60	Minor
b. Escape of gas around the primer cup (excluding a. above)	7	15	Minor
c. Escape of gas around the primer cup more than 50% of the periphery	12	25	Minor
	0	1	Major
d. Blown primer	0	1	Major
e. Dropped primer			
7. Case casualties 2/			
a. Longitudinal split	29	60	Minor
(1) Neck (I)	1	2	Major
(2) Body (K)	7	15	Minor
(3) Body (J)	0	1	Major
(4) To Head (L)	0	1	Major
(5) Through head (M)	27	65	Minor
(6) Shoulder (S)			
b. Circumferential rupture			
(1) Partial - neck, shoulder, or body (S, J, or K)	2	4	Minor
(2) Partial - case base (L)	0	1	Major
(3) Complete	0	0	Major
8. Case burn through	0	1	Major

NOTES:

1/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other tests exceed the acceptance number for the cumulative sample in Table VI. If the number of defects found in the first test exceeds the acceptance number but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of

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TABLE VI (Cont.)

double the quantities specified for the function and casualty test, shall be fired. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that the acceptance number for the cumulative sample is exceeded, the lot shall be rejected. If no acceptance number is listed for the cumulative sample, no second sample shall be permitted.

2/ For location of defects indicated in parenthesis, see drawing 7643674.

****

4.5.5 Packaging, packing, and marking inspection. Inspection for linking, packaging, packing and marking shall be IAW MIL-STD-644 as applies to the drawing or as required by the contract.

4.5.6 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the Inspection Reference column in the Qualification tables, First Article examination tables, and Quality Conformance Inspection Classification of Characteristics paragraphs. contractor shall submit inspection equipment designs for approval by the Government IAW the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.6 Methods of inspection.

4.6.1 Performance characteristics.

* 4.6.1.1 Primer sensitivity. Test IAW the procedure specified in Section 7-25 of TECP 700-700 Vol III using the rundown method with 50 primers at each height except that cartridge cases from the submitted lot shall be used using the equipment shown on EL8658292.

4.6.1.2 Residual stress.

* 4.6.1.2.1 Cartridges with brass cases. Test IAW paragraph 14 of Section 7-24 of TECP 700-700 Vol III using the equipment shown on EL8654131.

4.6.1.2.2 Cartridges with steel cases. Testing not required.

4.6.1.2.3 Cases of other materials. Test IAW procedures approved by the Government appropriate to the material(s) used to manufacture the cartridge.

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* 4.6.1.3 Waterproof. The cartridge shall be placed horizontally in a clear bell jar and covered with at least 26mm (1 inch) of water. The jar shall be evacuated to the specified differential pressure. Observation for air bubbles escaping from the cartridges shall be performed for 15 seconds minimum. Inspection equipment shown on EL8654133.

* 4.6.1.4 Function and casualty. The cartridges shall be tested in an M2 BMG with M19 BFA which is attached to the specified mount (M3 tripod, M4 pedestal or hard mount) and the M85 machine gun with M20 BFA attached to the hard mount. The linked cartridges shall be conditioned at 20C ±2C (68F ±3F) for not less than two hours, -20C ±5C (-4F ±9F) for not less than six hours, and 50C ±5C (122F ±9F) for four to twelve hours, -50C ±5C (-58F ±9F) for not less than six hours, and 70C ±5C (158F ±9F) for four to six hours. The weapons shall be at room temperature at the beginning of the test. The function and casualty test shall be conducted IAW TECP 700-700, Vol. III, Section 7-15, using the inspection equipment shown on EL8649406. Fire 100 round bursts. Record the cyclic rate for each burst. Cool the weapon to ambient temperature between each 100 round burst. Ammunition links shall not be reused in this test. observe for compliance with the appropriate firing defect accept-reject criteria table, Table II, Table III, or Table VI.

4.6.1.4.1 Cyclic rate. Measure and record the cyclic rates during function and casualty testing.

* 4.6.1.4.2 Noise. Fire ten ten round bursts of M33 Ball and ten ten round bursts of test cartridges placing the microphone of the noise level meter approximately .5 meters directly behind the weapon and 1 m to the left or 1 m to the right of the weapon. Inspection equipment shown on EL8649407 shall be used.

* 4.6.1.4.3 Screen perforation. The screen perforation test shall be conducted IAW TECP 700-700, Vol. III, Section 7-21 except that the target paper shall be 40 lb kraft paper, A-A-203. Inspection equipment shown on EL8649403 shall be used.

* 4.6.1.4.4 Muzzle flash. Firing in darkness with observers placed 100 meters down range behind a safe vision screen fire shorts bursts of approximately five rounds each with intervals of one minute between bursts. Record muzzle flashes observed. Inspection equipment shown on EL8649404 shall be used.

4.6.1.4.5 Breech flash. observe breech flash during function and casualty testing from a safe vantage point. Record any breech flashes observed and any breech flash that may be unsafe for the gunner.

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* 4.6.1.4.6 Fouling. Five 100 round bursts of the test cartridges shall be fired in full automatic action IAW TECP 700-700, Vol. III, Section 7-24, paragraph 7. The weapon shall be cooled to ambient temperature after each burst. Note that no reference rounds are to be used for this test using inspection equipment shown on EL8649405.

* 4.6.1.4.7 Attitude. One hundred cartridges shall be fired in a single burst from an M2 BMG with M19 BFA mounted on an M4 Pedestal Mount, 10885124, with Machine Gun Mount, 7046650, at an elevation of +85 degrees from the horizontal. One hundred cartridges shall be fired in a single burst from an M2 BMG with M19 BFA at an elevation of – 85 degrees from the horizontal.

* 4.6.1.4.8 Belt pull. The cartridges shall be fired in a continuous 100 round burst from the weapon with the 100 round belt of cartridges allowed to hang freely. The test shall be conducted IAW TECP 700-700 , Vol. III, Section 7-15, using the inspection equipment shown on EL8649406.

* 4.6.1.4.9 Hot chamber. A belt of 102 linked test cartridges shall be linked in the following manner: One hundred test cartridges followed by one dummy followed by one more test cartridge shall be linked up for a total belt of 102. The 102nd cartridge being the hot chamber effects test cartridge. Fire the 100 cartridges in a continuous burst in the weapon with BFA using the Inspection equipment shown on EL8649406. On misfiring on the 101 dummy cartridge, immediately recharge the weapon to bring the test cartridge into battery. Record the time if the test round fires. After 30 minutes remove the test cartridge and examine and record any damage to the test cartridge, the weapon, or to the barrel chamber.

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4.6.1.4.10 Chemical compatibility. The cartridges shall be dipped for one minute in the following compounds and then drained for 24 hours:

	Specification	Material
a.	MIL-L-46000	Lubricant, Semi-Fluid (LSA)
b.	MIL-L-14107	Lubricating Oil, (LAW)
c.	MIL-L-2104	Lubricating Oil, (ICE, T)
d.	MIL-L-46167	Lubricating Oil, (ICE, A)
e.	MIL-L-63460	Lubricant, (CLP)
f.	VV-L-800	Lubricating Oil, (PL-S)
g.	MIL-C-372	Bore Cleaner, (RBC)
h.	O-I-503	Insect Repellant
i.	P-D-680	Dry Cleaning Solvent (SD-1)

* The test cartridges shall then be wiped dry or blown off with compressed air and fired in ten round bursts in the weapon with BFA IAW TECP 700-700, Vol. III, Section 7-15, using the inspection equipment shown on EL8649406. Record the number of failures.

4.6.2 Environmental.

4.6.2.1 Sand and dust. The weapon with BFA assembled with a 200 round cartridge belt shall be placed in the sand and dust chamber. A sand and dust mixture as specified in TOP-3-2-045, Table 12, shall be dispersed on the weapon at a rate of 6.5 grams ± 2 grams per minute per square meter (100 ± 25 grains per minute per square meter) over the area concerned. The firing cycle shall begin one minute after the dispenser is turned on. The weapon shall be fired in 10 round bursts with a one minute interval between each burst. The test shall be repeated with Caliber .50 M33 Ball cartridges using the M2 BMG with the BFA removed. Malfunctions and cartridge failures shall be recorded.

* 4.6.2.2 Corrosion. The cartridges shall be subjected to a 5% salt spray solution for 48 hours. The test shall be performed IAW MIL-STD-810, Test 509. After completion of the test the cartridges shall be chambered and fired safely in the weapon with BFA, using the inspection equipment shown on EL8649406.

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* 4.6.2.3 Temperature/humidity cycling. The cartridges shall be conditioned IAW MIL-STD-810, Test 507, Procedure III. After completion of the conditioning the cartridges shall be fired in 25 round bursts in the weapon with BFA, using the inspection equipment shown on EL8649406. The cyclic rate shall be recorded.

4.6.2.4 Long term storage.

* 4.6.2.4.1 Long term high temperature storage. After storage for 30 days of continuous heating at 65C ±5C (150F ±9F) and 10 to 20% relative humidity, condition the cartridges at 20C ±2C (68F ±3F) for 24 hours. Upon completion of the conditioning the cartridges shall be fired in 10 round bursts in the weapon with BFA, using the inspection equipment shown on EL8649406. Record the cyclic rate. Any cartridge failures shall be recorded. If any cartridges fail the test at 65C, a retest shall be performed on cartridges stored at 50C ±5C (122F ±3F) and 10 to 20% relative humidity for 120 days.

* 4.6.2.4.2 Long term low temperature storage. After storage for 30 days at -40C ±5C (-40F ±9F), condition the cartridges at 20C ±2C (68F ±3F) for 24 hours. Upon completion of the conditioning the cartridges shall be fired in 10 round bursts in the weapon with BFA, using the inspection equipment shown on EL8649406. Record the cyclic rate. Any cartridge failures shall be recorded.

4.6.2.5 Rough handling.

* 4.6.2.5.1 Transportation-vibration and drop test. Four hundred cartridges packed in two wirebound boxes shall be subjected to the transportation-vibration test IAW TOP 1-2-601. Two hundred rounds shall be tested at -50C ±5C (-58F ±9F), and 200 rounds shall be tested at 70C ±5C (158F ±9F). Record any damage to the packaging and cartridges. After completion of this test, 200 each of the same cartridges shall be subjected to the 1.5 meter, loose cargo, and 2.1 meter drop tests IAW TOP 4-2–602. After completion of these tests the cartridges shall be fired in 10 round bursts in the weapon with BFA, using the inspection equipment shown on EL8649406. Record the cyclic rate. Any cartridge failures shall be recorded.

4.6.2.5.2 12 meter drop test. Four hundred cartridges Packed in two wirebound boxes shall be subjected to the 12 meter (40 ft) drop test IAW TOP 4-2-601. Two hundred rounds shall be tested at -50C ±5C (-58F ±9F), and 200 rounds shall be tested at 70C ±5C (158F ±9F). Record any damage to the packaging and cartridges.

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4.6.2.5.3 Toxic fumes. The cartridges shall be fired in 10 round bursts in the weapon with BFA. The muzzle of the weapon shall protrude from a 2130mm x 2130mm x 2130mm chamber. The air within the chamber shall be continuously monitored for carbon monoxide, ammonia, nitrogen dioxide, and sulfur dioxide. Three trials of 100 rounds each shall be performed. The test shall be repeated with Caliber .50 M33 Ball cartridges using the M2 BMG with the BFA removed.

4.6.3 Safety.

4.6.3.1 Safety hazard classification. The test shall be conducted by the contractor at a facility of his choosing at his expense. The Government shall reserve the right to observe all testing. The tests shall be conducted IAW TB 700-2 with the following differences: only one outer packing unit shall be placed on the stand and the stand shall be made of metal instead of wood on the external stack test (paragraph 5-3g).

4.6.3.2 Explosive sensitivity. Upon design submission the U.S. Army Safety Office shall determine which materials utilized in the cartridge require sensitivity evaluation testing by the contractor. The sensitivity tests required shall be as follows:

	Test	Test Document	Test Procedure/ Paragraph
a.	Friction test - apparatus and comparison values	JSSPM	5.3
b.	Impact test - apparatus and comparison values	JSSPM/ TB 700-2	4.2 5.2i
c.	Electrostatic discharge test - apparatus and comparison values	JSSPM	4.2
d.	Auto ignition temperature	JSSPM	5.9
e.	Explosive temperature (5 seconds)	MIL-STD-650	506.1
f.	Detonation test	TB 700-2	5.2f
g.	Ignition and unconfined burning test	TB 700-2	5.2g
h.	Thermal stability test	TB 700-2	5.2h
i.	Card gap test	TB 700-2	5.2j

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4.6.4 Test validity. If for any reason the test agency considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and new test shall be performed with additional samples.

4.6.5 Defect penalty. In any ballistic test in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance sample shall be penalized for such defects IAW the appropriate firing defects table. Firing defects are shown on tables Table II, Table IV, or Table VI.

5. PACKAGING

5.1 Packaging level A. The contractor shall package the linked blank ammunition in the M2A1 metal ammunition container. 7553296 IAW 12576457. The container shall contain 100 cartridges when linked with the M9 link and shall contain 85 cartridges when linked with the M15A2 link. The contractor shall devise suitable internal dunnage for the M2A1 container to insure that the packaged ammunition is protected from any damage when subjected to the rough handling tests in 3.5.5. Shipment shall be made IAW CFR TITLE 49 and MIL-STD-129.

5.2 Packaging level A. The loaded M2A1 metal containers shall be placed in the wood wirebound box, 7553347, in pairs IAW the packaging instructions on drawing 12576456. Shipment shall be made IAW CFR TITLE 49 and MIL-STD-129.

5.3 Marking. Containers shall be marked IAW MIL-STD-129 and CFR 49. Drawing 12576456 shall be used for marking format but not content. A Performance Oriented Packaging symbol shall be provided by the Government after successful completion of Performance oriented Packaging Tests (see 5.6) and documentation in a test report provided to the Government. Proper Shipping Name and UN Identification Marking shall be provided by the Government on successful completion of Hazard Classification Testing (see 4.6.3). All other marking shall be as specified in the contract.

5.4 Palletization. Packaged wood wirebound boxes shall be palletized IAW the quantities and procedures shown on drawing 19-48-4116/14.

5.5 Documentation of technical data. All information needed to package, pack, and palletize the ammunition for both link configurations shall be provided to the Government in Level 3 format IAW DOD-STD-100.

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5.6 Performance Oriented Packaging Testing and Documentation. The contractor shall conduct, at his expense, Performance oriented Packaging Tests IAW the test procedures in CFR 49. Testing shall be conducted at a DOT certified test facility using live ammunition. Testing shall be conducted on packs with ammunition linked with both M9 and M15A2 links. The US Government shall be advised 45 days prior to scheduled testing so what the Government may witness tests at the discretion of the Government. Test procedures and results shall be documented in a test report and forwarded to the Government for review. This testing is in addition to the rough handling testing specified In 3.5.5.

6. NOTES

(This section contains information of a general or explanatory nature that may be be helpful, but is not mandatory.)

6.1 Intended use. The cartridges covered by this specification are intended for use in Caliber .50, M2 Browning Machine Gun with the M19 Blank Firing Attachment, weapon system and the Caliber .50, M85 Machine Gun with the M20 Blank Firing Attachment, weapon system.

6-2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Certificate of conformance for each lot or shipment of cartridges.

g. Type and level of packing for the cartridges.

h. Provisions for the submission of Inspection Equipment Designs (DI-R-100540) (see 6.3).

i. Provisions for the submission of acceptance inspection reports containing the final inspection results for each lot of ammunition presented to the Government (see 6.4).

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6.3 Submission of contractor inspection equipment designs for approval. (See MIL-A-48078). Submit copies of designs as required to: Commander, ARDEC, ATTN: SMCAR-QAF-I (D), Picatinny Arsenal, N.J. 07806-5000. This address will be specified on the Contract Data Requirements List (CDRL), DD Form 1423 in the contract.

6.4 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the US Army AMCCOM, one copy of all ballistic data and the Ammunition Data Card for each lot shall be forwarded to: Commander, ARDEC, ATTN: SMCAR-QAF-S (D), Picatinny Arsenal, N.J. 07806-5000.

6.5 Hazard notice. The cartridge described herein and some of the cartridge components are flammable and explosive and, therefore, present hazards in manufacture, handling, storage, and shipment. the contractor shall -recognize these hazards and shall take appropriate measures to prevent fire, explosion, adverse environmental, rough handling, corrosive atmosphere, and electrically induced incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.6 Drawings. Drawings listed in Section 2 of this document under the heading US Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, U.S. Army Armament Research and Development Command (ARRADCOM), or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Definitions.

6.7.1 Reliability. The number of successes (rounds chambered, fired, and ejected) divided by the number tested. occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall this type of failure be recorded as a defect chargeable to the ammunition.

6.8 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, any proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

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6.9 Subject term (key word) listing.

M1A1
M2 Browning Machine Gun
M85 Machine Gun.
Machine Gun.
Munitions
Small Arms Ammunition

* 6.10 Changes from Previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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3. DOCUMENT TITLE: CARTRIDGE, CALIBER .50, BLANK, XM928, PERFORMANCE SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
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